QR Code Generator Using Python

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An assignment submitted in partial fulfillment of the requirements for MSCS-633 as part of the degree Master of Science

In

Advance Artificial Intelligence

School of Computing and Information Sciences

University of the Cumberlands

MSCS-633: Advance Artificial Intelligence

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October 29, 2026

Fall 2026

1. Application Design and Implementation

The application was implemented using separate functions to keep the code organized and easier to maintain. The implementation consists of three main functions:

1. URL Validation

The validate_url() function accepts the URL entered by the user and checks whether it is valid. If the user enters a website address without specifying a protocol, the application automatically adds HTTPS. It also checks for an empty input, unsupported protocols, and invalid host information.

2. QR Code Generation

The generate_qr_code() function uses the Python qrcode library to generate the QR code.

The following settings are used:

- Automatic QR code version selection based on the input data.
- High error correction level.
- Black QR code pattern with a white background.
- Four-module border around the QR code.
- PNG image output.

The generated image is saved as generated_qr.png in the project directory.

3. Main Application

The main() function controls the overall execution. It displays instructions, accepts the URL from the user, calls the validation and QR generation functions, and displays a success message after the image is saved.

The application also includes exception handling to display meaningful messages when the user enters invalid information or when an error occurs while saving the image.

2. Application Execution

The application can be executed from the project directory using the following steps.

Step 1: Create a virtual environment

```
python3 -m venv .venv
```

Step 2: Activate the virtual environment

```
source .venv/bin/activate
```

Step 3: Install the required dependencies

```
python -m pip install -r requirements.txt
```

Step 4: Check the Python source code

```
python -m py_compile qr_generator.py
```

Step 5: Run the application

```
python qr_generator.py
```

When prompted, enter a website URL. For example:

https://www.bioxsystems.com/

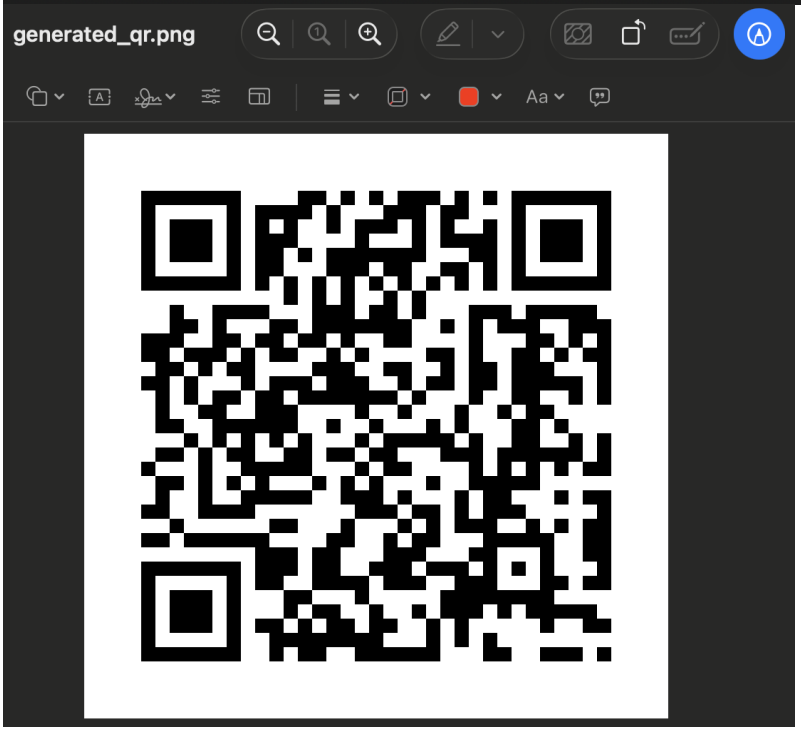
After successful execution, the application generates a QR code image named generated_qr.png.

```
(.venv) sadhanakhadka@Mac-479 QR-Code-Generator % python3 qr_generator.py

==================================
       QR CODE GENERATOR
==================================
Enter a website URL to generate its QR code.
Example: https://www.bioxsystems.com/

Enter URL: https://www.bioxsystems.com/

QR code generated successfully!
Website URL : https://www.bioxsystems.com/
Output file : /Users/sadhanakhadka/Documents/SabinUC/AI/QR-Code-Generator/generated_qr.png
```



3. GitHub Repository

GitHub Repository URL: https://github.com/rbsabin/QR-Code-Generator.git

References

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